

A deep-learning approach to predict airline passenger satisfaction

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Dataset Characteristics

- 103,904 observations in training set
- 25,976 observations in test set
- 23 features including demographics (Age, Gender)
- Flight details (Distance, Class, Departure and Arrival time)
- 14 service ratings on a 0–5 Likert scale
- Airline satisfaction level (Satisfaction, neutral or dissatisfaction)

Research issue

- Very large data where we want to predict a categorical (Satisfied or not) variable
- We managed to apply some unsupervised models looking for **patterns** and several supervised learning models to predict the **labels**.
- The task is a **classification** one, so we can apply for example logistic regression
- The dataset is available online on the kaggle platform

Data Cleaning & Transformation



Cleaning

Dropped 'id', handled 393 null values in 'Arrival Delay' by listwise deletion.



Encoding

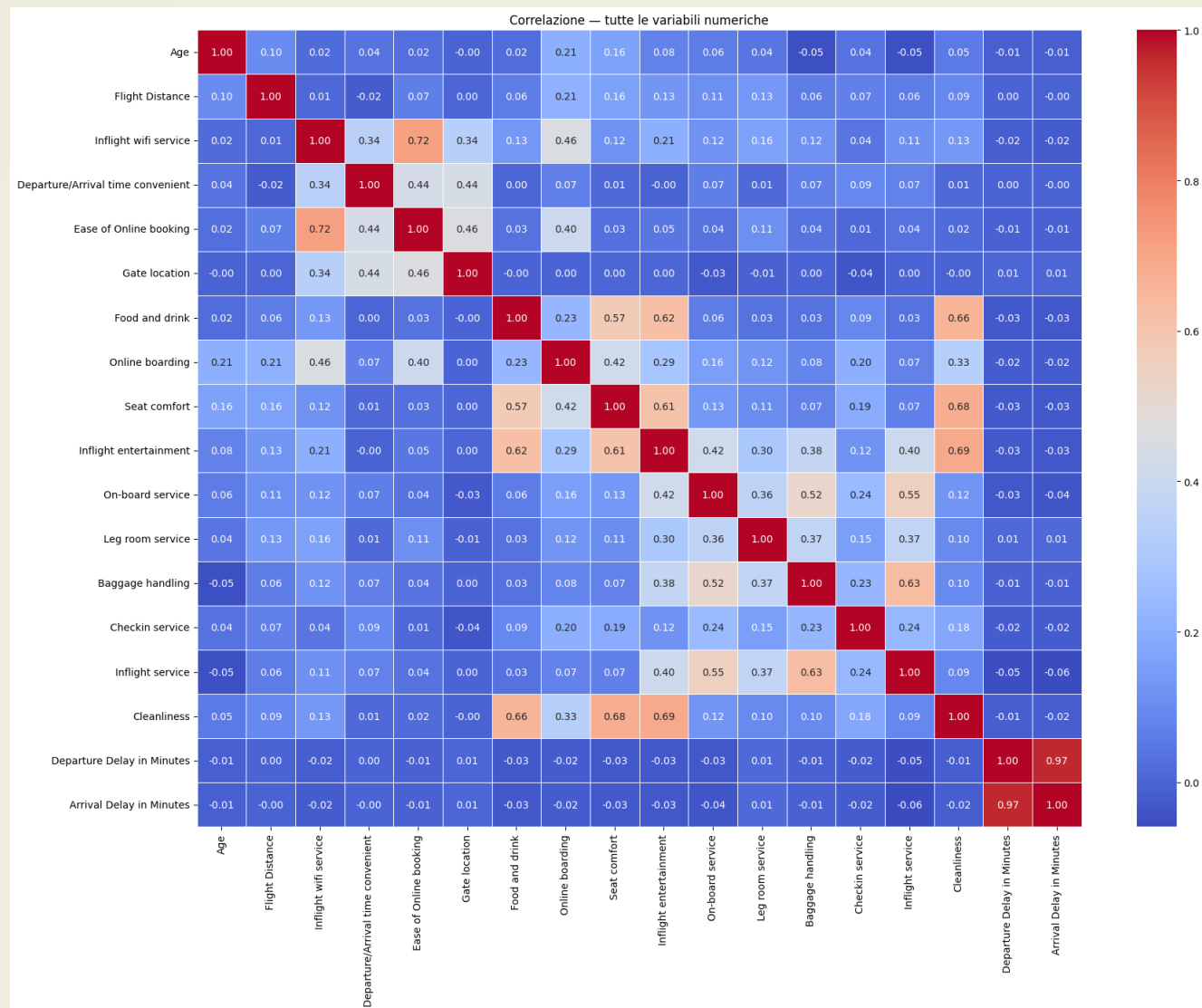
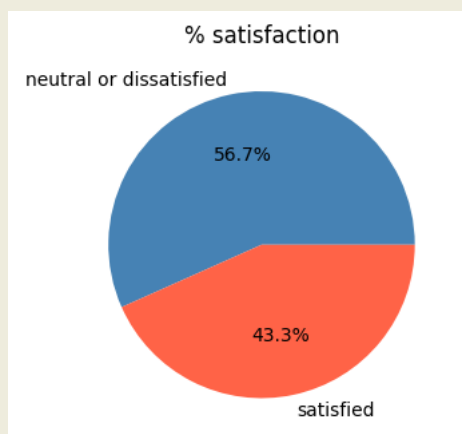
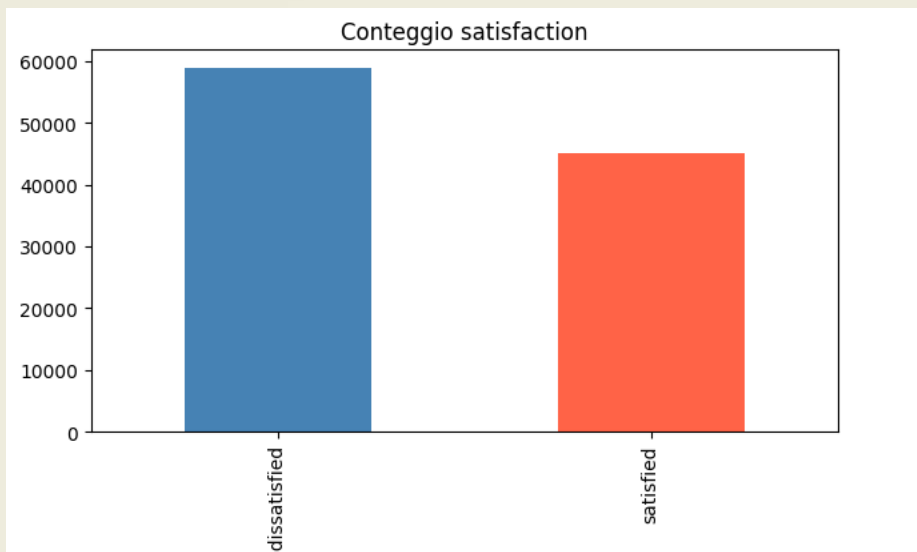
Label Encoding for Gender, Customer Type, and Travel Type. Manual mapping for 'Class' (0-2).



Scaling

Scaled features by type: log1p + RobustScaler for skewed variables, MinMaxScaler for ratings, passthrough for binary features.

Exploratory Data analysis

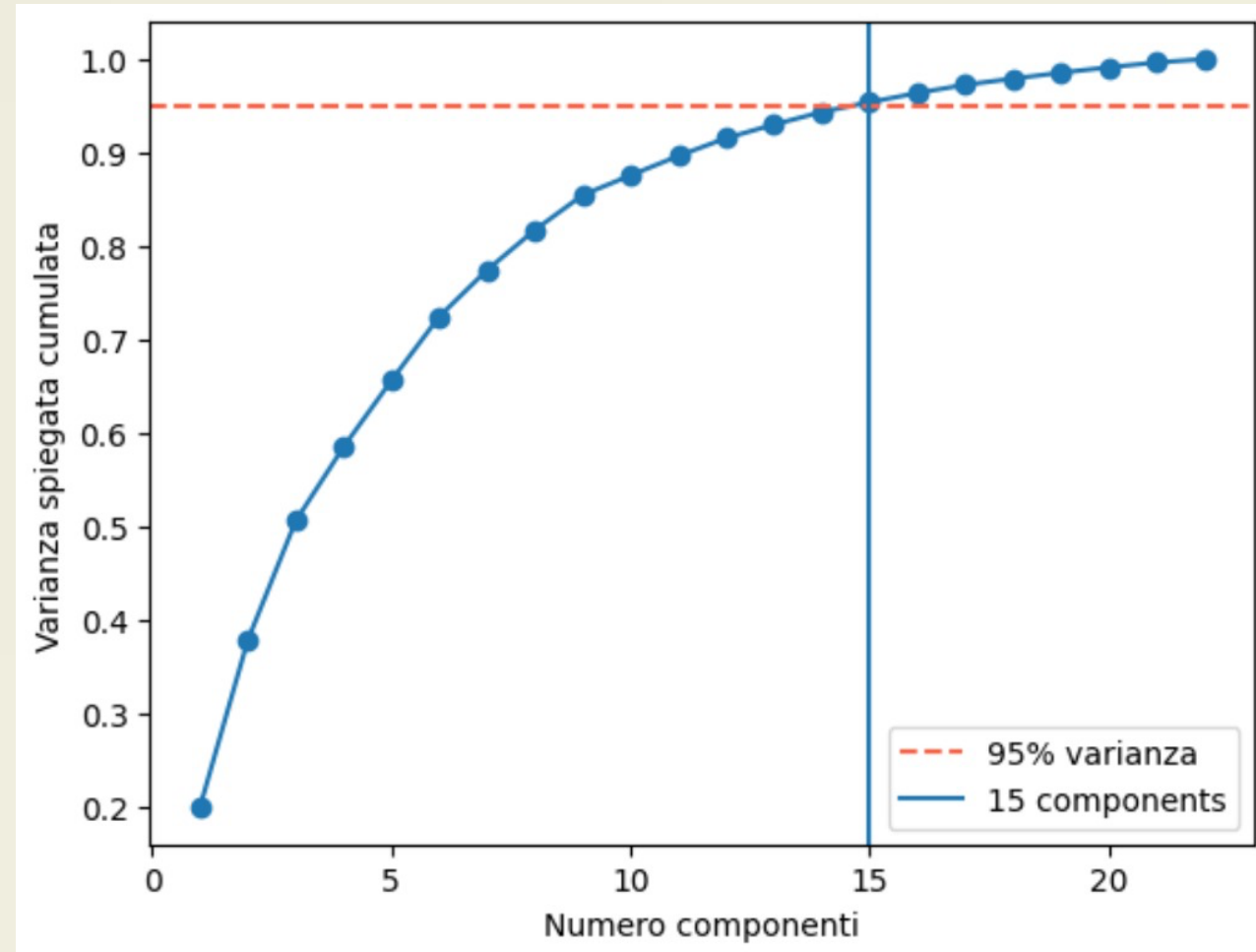


Dimensionality Reduction (PCA)

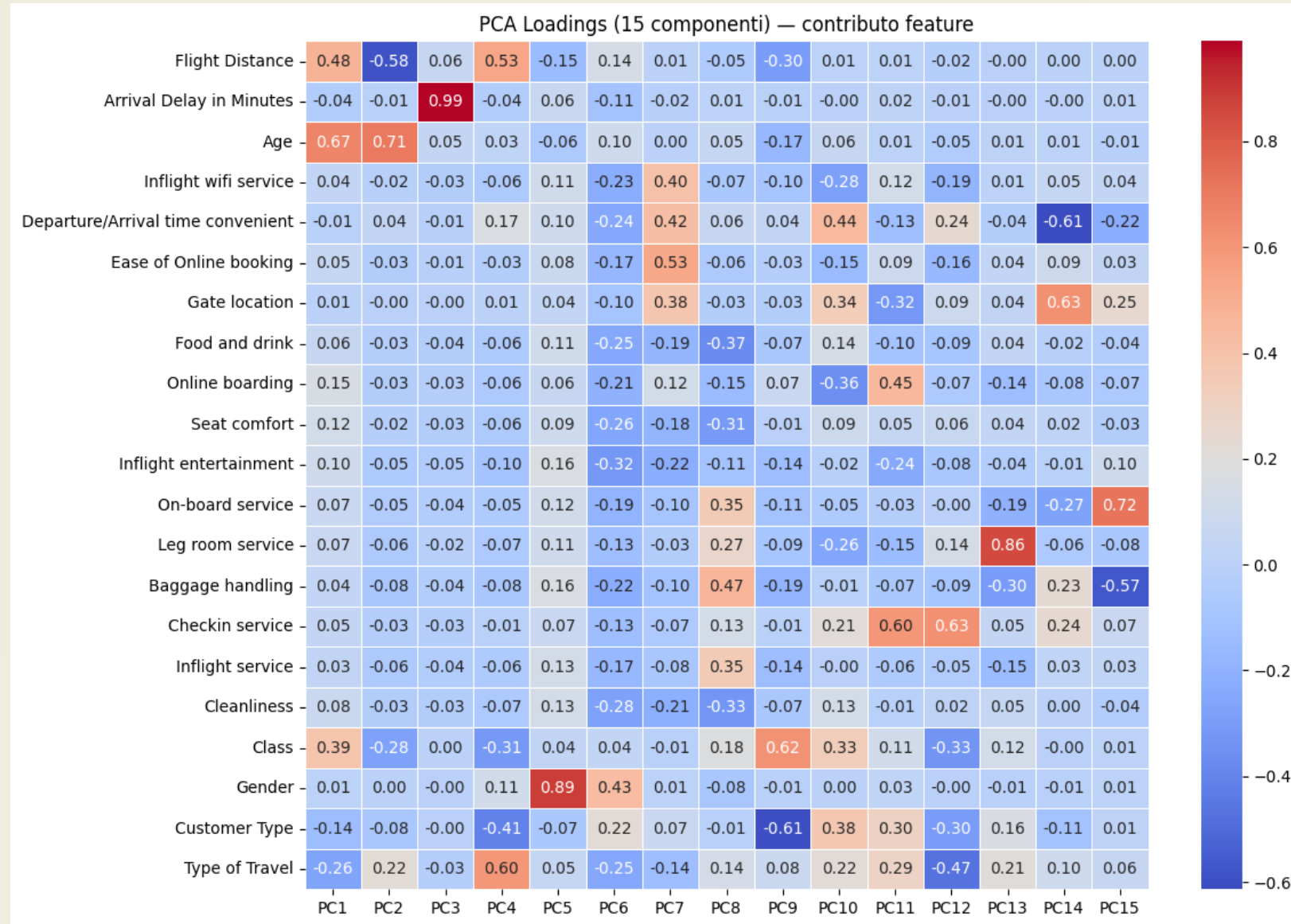
Variance Analysis

Standard Principal Component Analysis was conducted on the scaled features to reduce noise and multicollinearity.

- ✓ 15 components explain 95% of total variance.



Principal Components features explanation



Agglomerative Hierarchical Clustering

Clustering Methodology

Distance & Linkage

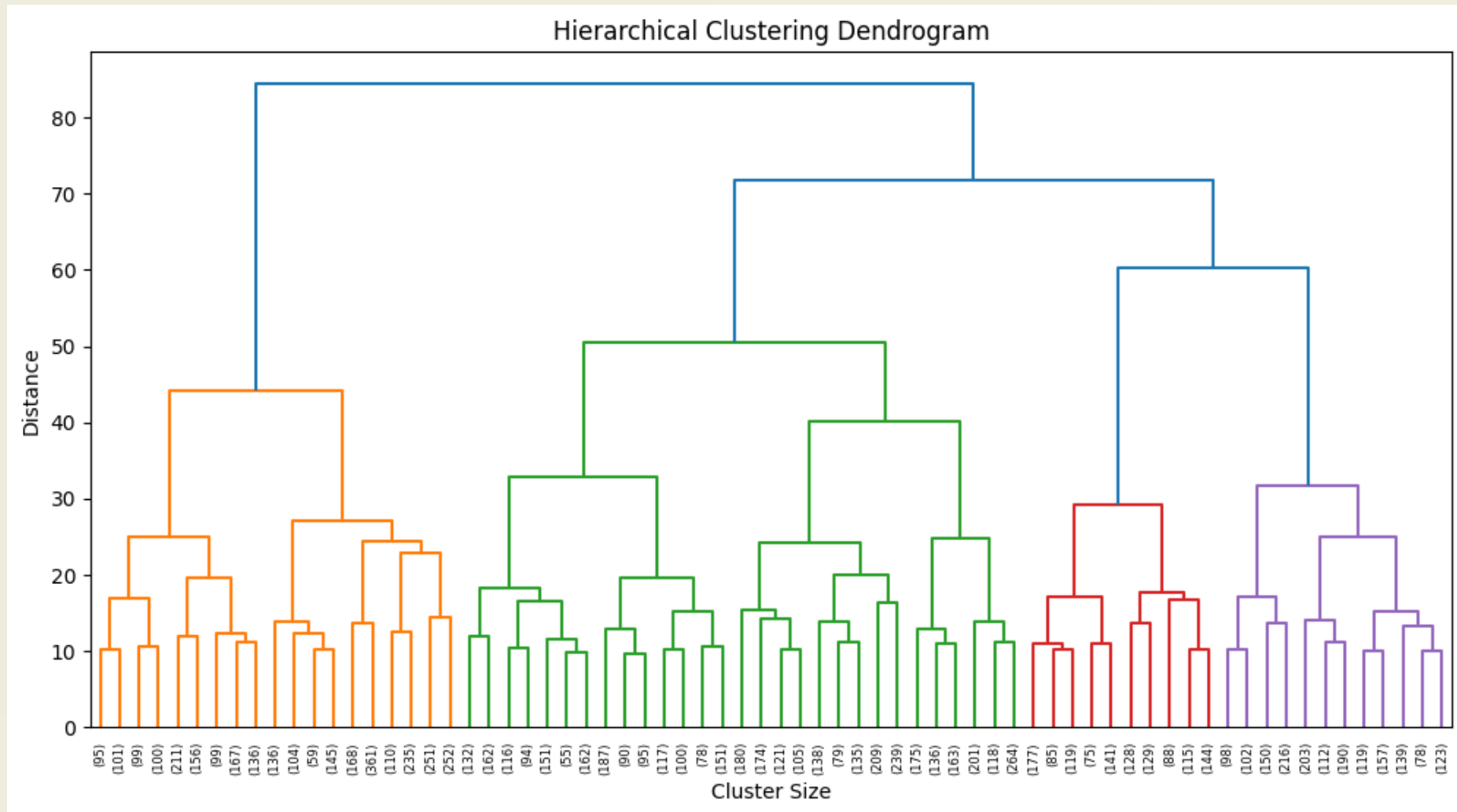
We utilized **Ward's Method**, which minimizes the total within-cluster variance. Euclidean distance was chosen as the metric for the scaled continuous and ordinal space.

Computational Sampling

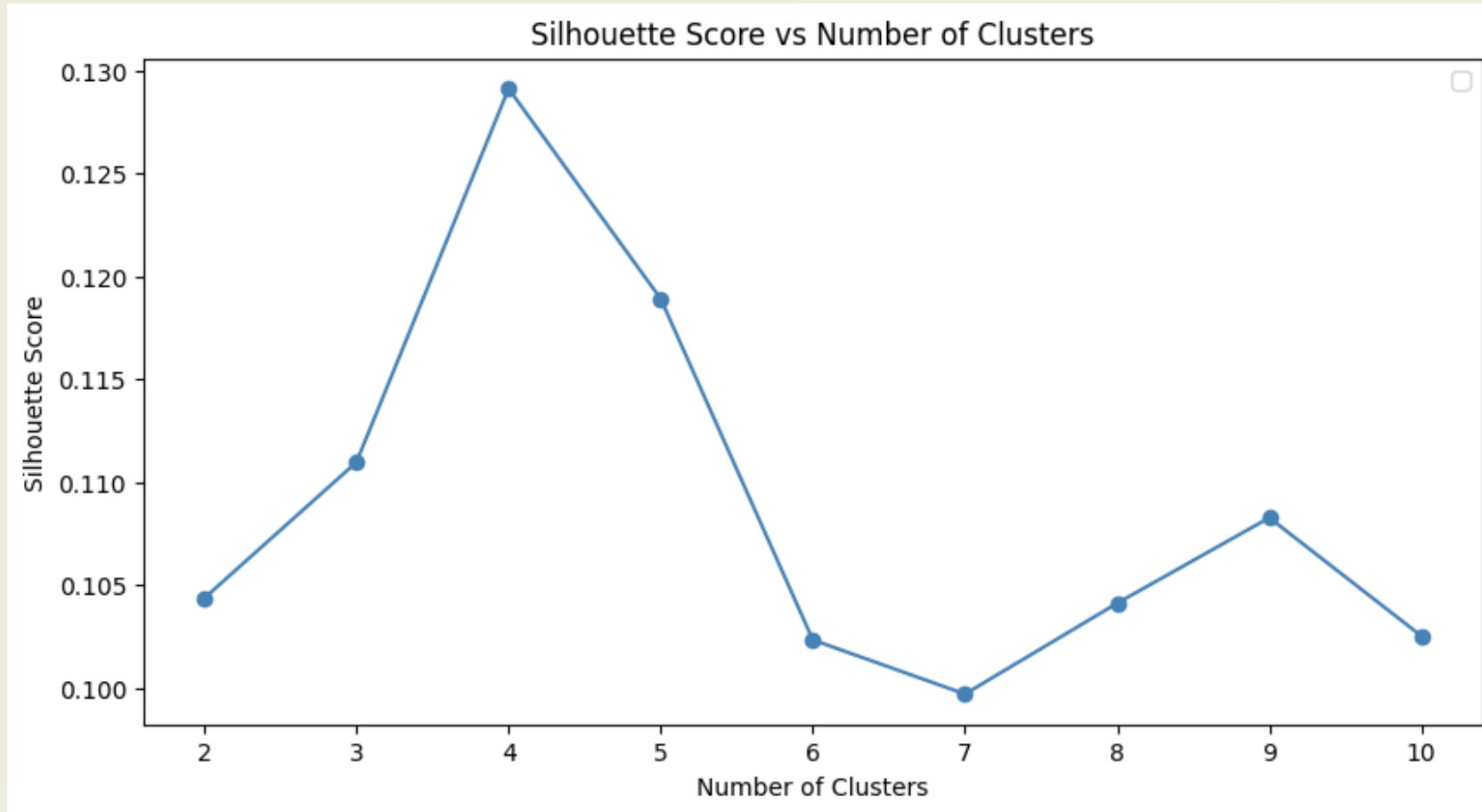
Due to the $O(n^2)$ complexity of hierarchical clustering, the algorithm was executed on a random sample of **10,000 observations** to ensure dendrogram legibility.

Hierarchical Agglomerative Clustering

After PCA (95% Variance)



Performance Metrics: Silhouette Score



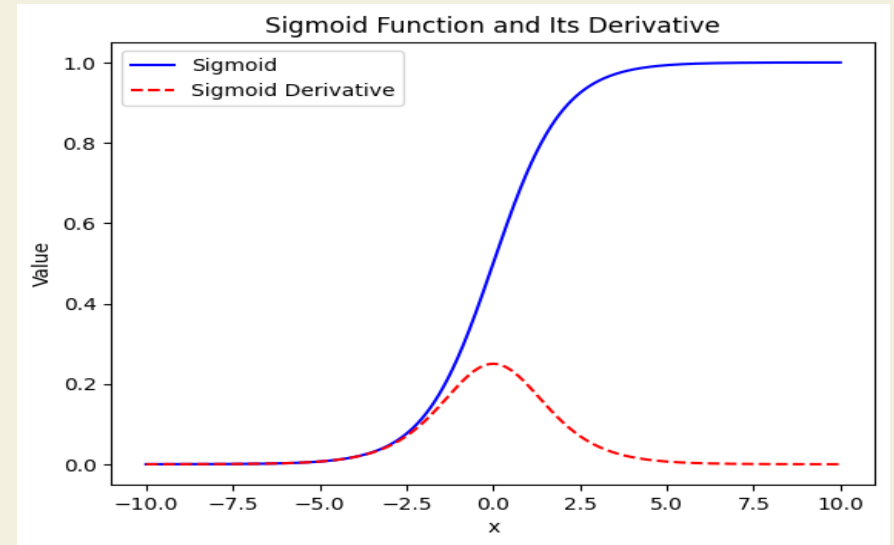


Supervised Learning Methods

Elastic Net

$$\log \left[\frac{p(x)}{1-p(x)} \right] = w \cdot x + b \quad \longrightarrow \quad p(x) = \frac{1}{1+e^{-(w \cdot x + b)}}$$

Probability that the input x belongs to Class $y=1$ (satisfied)



$$\log(L_{\text{total}}) = \sum_{i=1}^n \left[y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i)) \right] - \lambda \left(0.5 \sum_{j=1}^d |w_j| + 0.25 \sum_{j=1}^d w_j^2 \right)$$

The Log-Likelihood is the objective function that measures how well the $\mathbf{w}$ and $\mathbf{b}$ explain the observed data

$$\frac{\partial \log(L_{\text{total}})}{\partial w_j} = \sum_{i=1}^n \underbrace{(y_i - p(x_i))}_{\text{Errore}} x_{ij} - \lambda \left(0.5 \cdot \text{sign}(w_j) + 0.5 \cdot w_j \right)$$

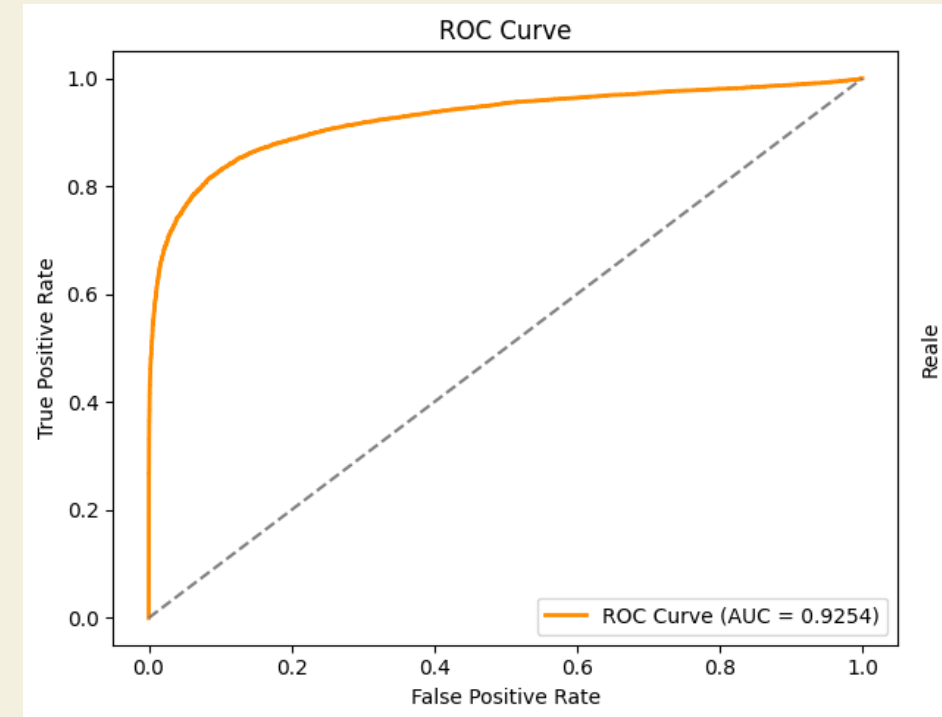
Gradient ascent of the log-likelihood function to find the best $\mathbf{w}$ and $\mathbf{b}$

Elastic Net

Feature	Log_Odds_Coef	Odds_Ratio
Online boarding	2.320206	10.177771
Checkin service	1.922149	6.835630
On-board service	1.384559	3.993063
Inflight wifi service	1.231176	3.425257
Leg room service	1.061677	2.891216
Inflight entertainment	0.862642	2.369412
Seat comfort	0.827714	2.288083
Ease of Online booking	0.812047	2.252515
Class	0.728949	2.072902
Inflight service	0.621005	1.860797

Interpretation Example:

For every 1 standard deviation increase in 'Online boarding', the odds of the outcome being 'Satisfied' are multiplied by 10.18.



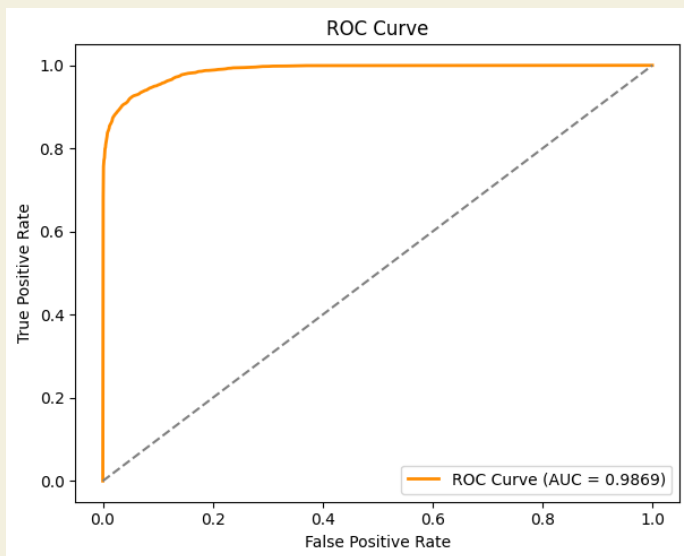
Decision Tree

```
param_dist = {  
    'criterion': ['gini', 'entropy'],  
    'max_depth': randint(5, 50),  
    'min_samples_split': randint(2, 150),  
    'min_samples_leaf': randint(1, 100),  
    'min_weight_fraction_leaf': uniform(0.0, 0.1)  
}  
  
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)  
  
random_search = RandomizedSearchCV(  
    estimator=DecisionTreeClassifier(random_state=42),  
    param_distributions=param_dist,  
    n_iter=50,  
    cv=skf,  
    scoring='f1',  
    n_jobs=1,  
    random_state=10,  
    verbose=0  
)
```



Hyperparameters{
Criterion: 'gini'
'max_depth': 21
'min_saples_leaf': 75
'min_samples_split' = 134
}

Decision Tree



===== VALUTAZIONE: Decision Tree =====
ROC-AUC Score: 0.9869 (Indipendente dalla soglia)

--- SOGLIA DEFAULT (0.5000) ---

Accuracy: 0.9373

F1 Score: 0.9265

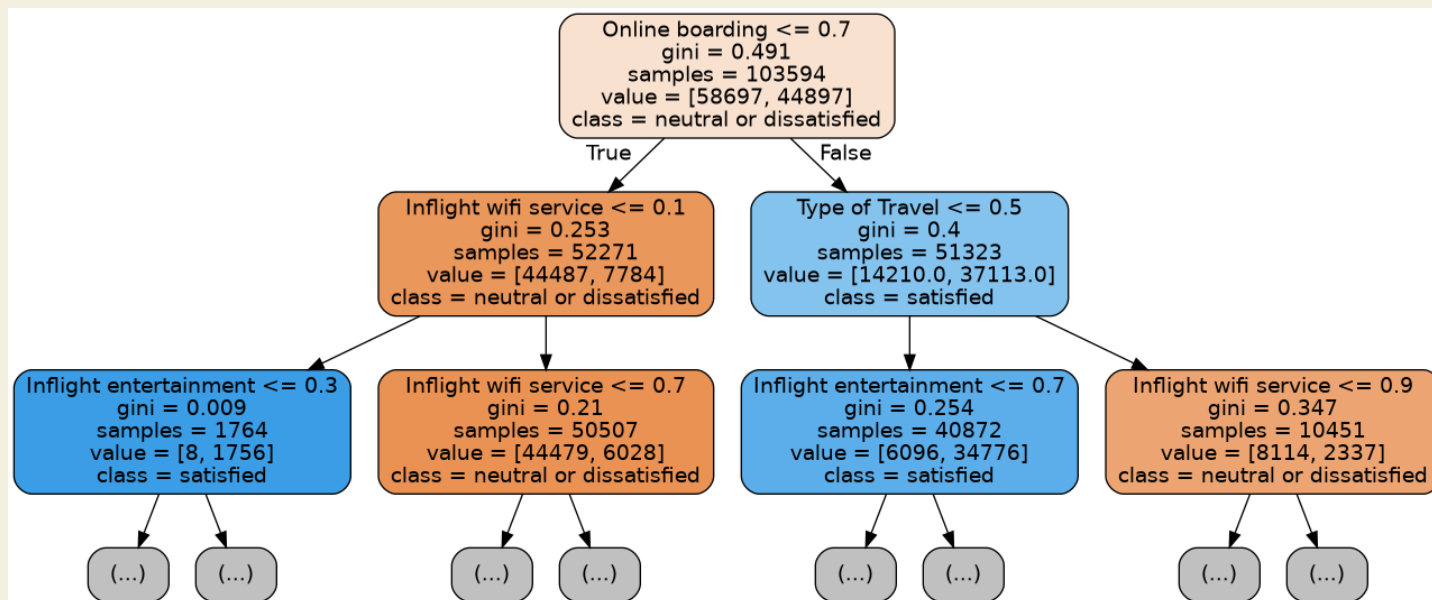
--- SOGLIA OTTIMALE (0.4424) ---

Accuracy: 0.9371

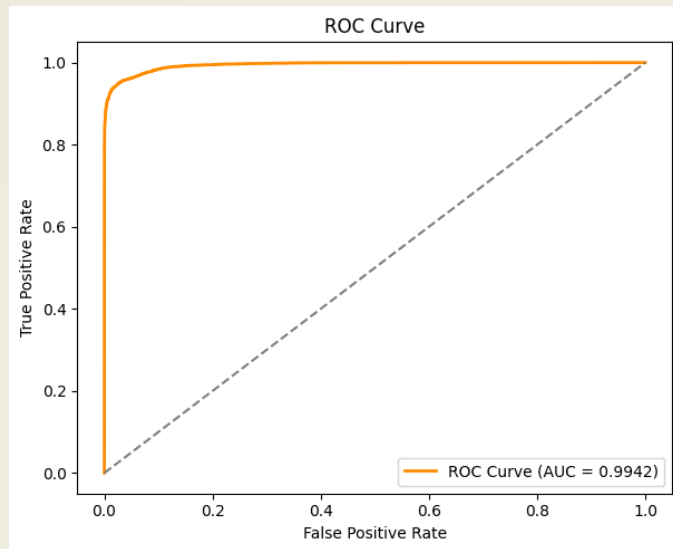
F1 Score: 0.9280

Report Completo (Soglia Ottimale):

	precision	recall	f1-score	support
not satisfied	0.94	0.95	0.94	14528
satisfied	0.93	0.92	0.93	11365
accuracy			0.94	25893
macro avg	0.94	0.94	0.94	25893
weighted avg	0.94	0.94	0.94	25893



Random Forest



```
===== VALUTAZIONE: Random Forest =====
ROC-AUC Score: 0.9942 (Indipendente dalla soglia)

--- SOGLIA DEFAULT (0.5000) ---
Accuracy: 0.9631
F1 Score: 0.9574

--- SOGLIA OTTIMALE (0.4617) ---
Accuracy: 0.9630
F1 Score: 0.9576

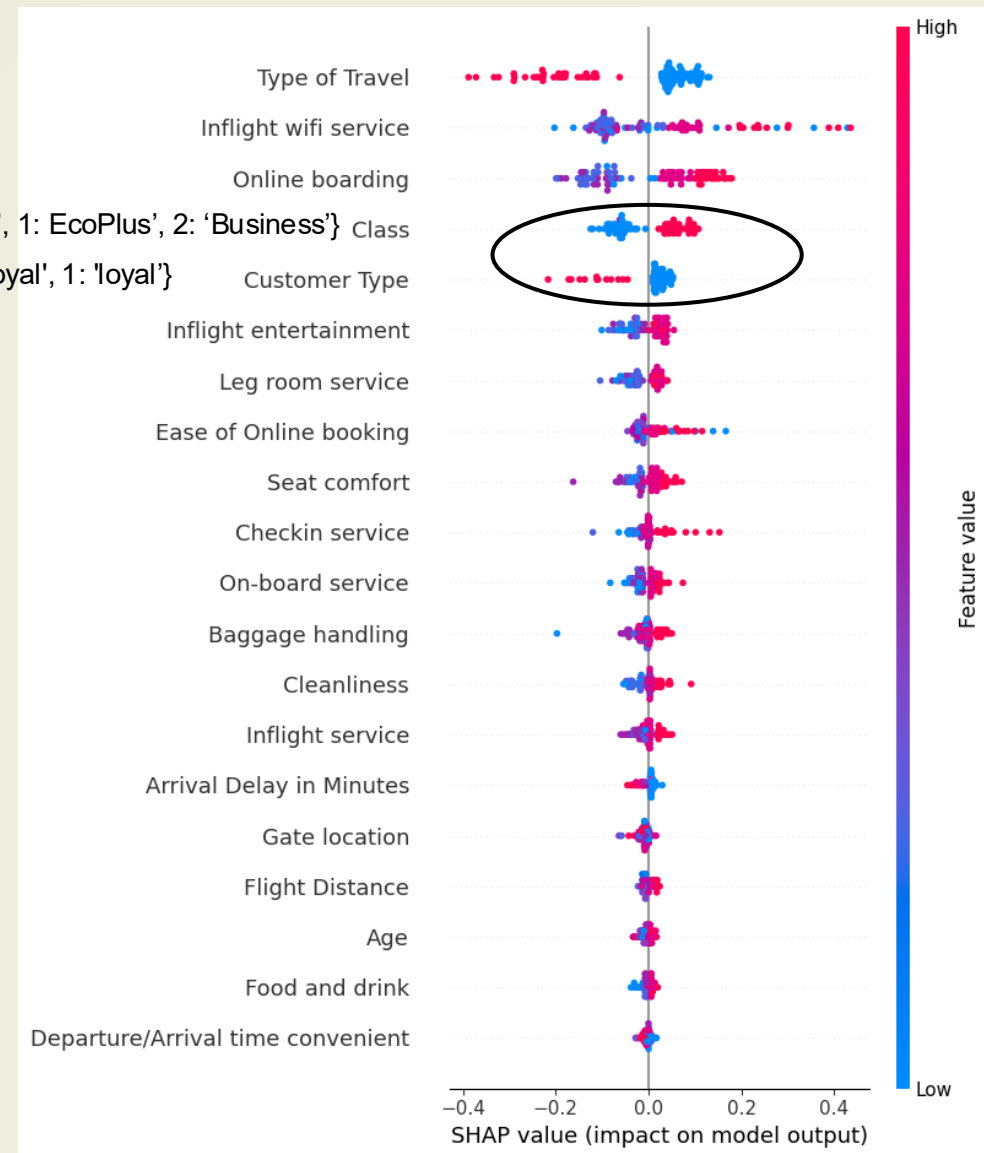
Report Completo (Soglia Ottimale):
      precision    recall  f1-score   support

not satisfied      0.96      0.97      0.97     14528
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   accuracy
macro avg      0.96      0.96      0.96     25893
weighted avg    0.96      0.96      0.96     25893
```

{0: 'Eco', 1: EcoPlus', 2: 'Business'} Class

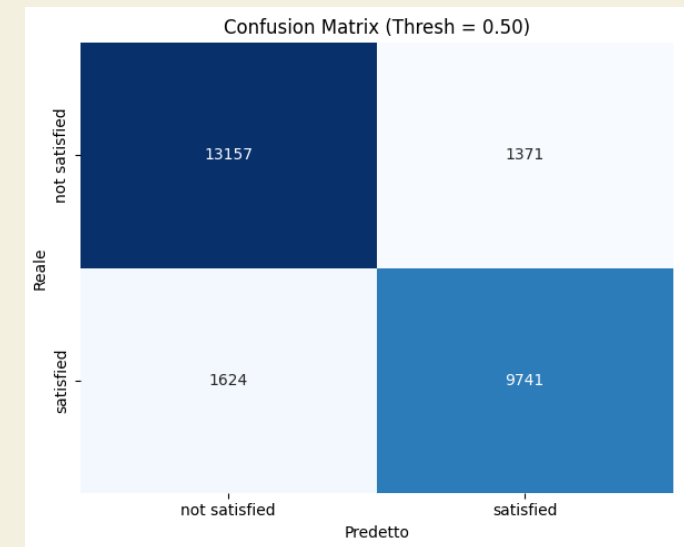
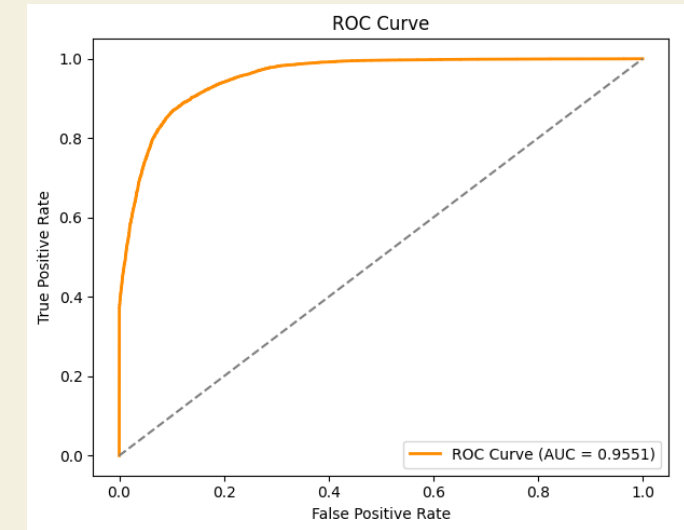
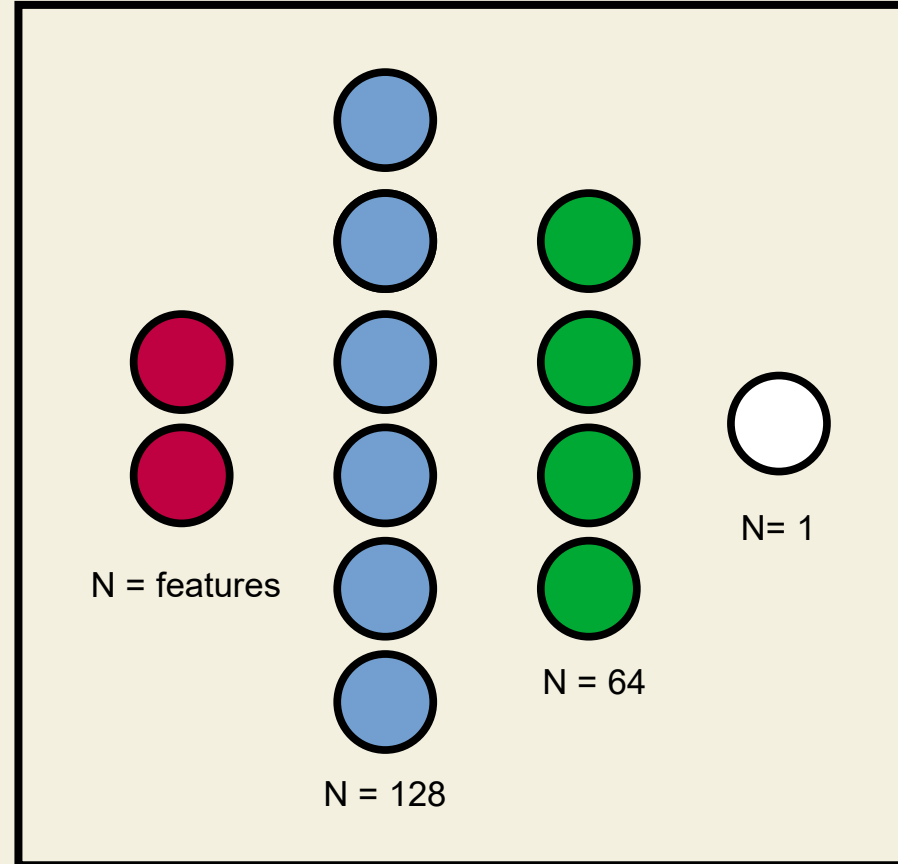
{0: 'disloyal', 1: 'loyal'} Customer Type



Let's make it realistic

We decided to **drop** also:

- 1) 'Inflight wifi service',
- 2) 'Food and drink',
- 3) 'Seat comfort',
- 4) 'Inflight entertainment',
- 5) 'On-board service',
- 6) 'Leg room service',
- 7) 'Inflight service',
- 8) 'Cleanliness',
- 9) 'Checkin service',
- 10) 'Baggage handling'



Thank you for your attention!

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